

Wenyi Liu

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Education

The University of Hong Kong, Ph.D. in Robotics Mar 2023 - Present

- **Supervisor:** Prof. Fu Zhang
- **Laboratory:** Mechatronics and Robotic Systems (MaRS) Laboratory
- **Scholarship:** The Arthur and Louise May Memorial Fund Scholarship (2024-2025)
- **Specialization:** Robot Navigation, Swarm Robotics, Planning and Control, Uncrewed Aerial Vehicle (UAV), Aerial Manipulation, Embodied AI, LiDAR-based Perception
- **Skills:** Linux, C++ , ROS

Harbin Institute of Technology, Shenzhen, B.Eng. in Communication Engineering Sep 2018 – Jun 2022

- **Scholarship:** Outstanding Student Scholarship (2018-2021)
- **Thesis:** Trajectory Planning for High-Speed Mobile Takeoff and Landing of UAVs, Outstanding Thesis Award
- **Specialization:** Analog and Digital Circuits, Signals and Systems, Robotics, Deep Learning
- **Skills:** Mechanical Design, Embedded Hardware, Embedded Software, Control Algorithm Design, Python

Publications

- [1] **W. Liu**, Y. Ren, R. Guo, V. W. W. Kong, A. S. P. Hung, F. Zhu, Y. Cai, H. Wu, Y. Zou, and F. Zhang. "Slope inspection under dense vegetation using LiDAR-based quadrotors" in **Nature Communications (NC 2025)**. [Paper] [Code] [Video]
- [2] **W. Liu***, H. Wu*, L. Shi, F. Zhu, Y. Zou, F. Kong, and F. Zhang. "LiDAR-Based Quadrotor Autonomous Inspection System in Cluttered Environments" in **IEEE Transactions on Field Robotics (T-FR 2025)**. [Paper] [Video]
- [3] **W. Liu***, Y. Ren*, and F. Zhang. "Integrated planning and control for quadrotor navigation in presence of suddenly appearing objects and disturbances" in **IEEE Robotics and Automation Letters (RA-L 2023)**. [Paper] [Code] [Video]
- [4] **W. Liu***, L. Shi*, Y. Geng, Y. Zhu, X. Han, Y. Jiang, H. Wu, Y. Zou, L. Yin, N. Chen, L. Jia, Y. Qin, Z. Wang, Q. Wang, F. Miao, C. Zhao, and F. Zhang. "DIRECT: A digital reflex for UAV navigation in clutter under uncertain disturbances" (Under Review). [Video]
- [5] L. Shi*, **W. Liu***, X. Han, Y. Ren, L. Yin, F. Zhu, N. Chen, and F. Zhang. "Traversing the Untraversable: Enabling UAV Navigation in the Extremely Dense Wild by Contact-driven Adaptation" in **IEEE Transactions on Field Robotics (T-FR 2026)**. [Paper] [Video]
- [6] H. Wu*, **W. Liu***, Y. Ren, Z. Liu, H. Wei, F. Zhu, H. Li, and F. Zhang. "Flying through cluttered and dynamic environments with LiDAR" in **Advanced Intelligent Systems (AIS)** (Under Review). [Paper]
- [7] Z. Yu, Z. Qiao, **W. Liu**, H. Yin, and S. Shen. "SLIM: Scalable and Lightweight LiDAR Mapping in Urban Environments" in **IEEE Transactions on Robotics (T-RO 2025)**. [Paper] [Code] [Video]
- [8] F. Zhu*, Y. Ren*, L. Yin, F. Kong, Q. Liu, R. Xue, **W. Liu**, Y. Cai, G. Lu, H. Li and F. Zhang. "Swarm-LIO2: Decentralized Efficient LiDAR-Inertial Odometry for Aerial Swarm Systems" in **IEEE Transactions on Robotics (T-RO 2024)**. [Paper] [Code] [Video]
- [9] L. Yin, Y. Ren, F. Zhu, L. Shi, F. Kong, B. Tang, **W. Liu**, X. Lyu and F. Zhang. "Visibility-aware Cooperative Aerial Tracking with Decentralized LiDAR-based Swarms" in **IEEE Transactions on Robotics (T-RO)** (Under Review).
- [10] B. Tang, Y. Ren, Y. Cai, F. Kong, **W. Liu**, F. Zhu, L. Yin, L. Shi and F. Zhang. "Memory-Efficient Boundary Map for Large-Scale Occupancy Grid Mapping" in **International Journal of Robotics Research (IJRR 2026)**. [Paper] [Code] [Video].

- [11] Z. Liu, H. Li, C. Yuan, X. Liu, J. Lin, R. Li, C. Zheng, B. Zhou, **W. Liu**, and F. Zhang. "Voxel-SLAM: A Complete, Accurate, and Versatile Light Detection and Ranging-Inertial Simultaneous Localization and Mapping System" in **Advanced Intelligent Systems (AIS 2026)**. [Paper] [Code] [Video]
- [12] Y. Ren*, F. Zhu*, **W. Liu**, Z. Wang, F. Gao, and F. Zhang. "Bubble Planner: Planning High-speed Smooth Quadrotor Trajectories using Receding Corridors" in **IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2022)**. [Paper] [Video]

Awards

Best Contribution Award in the Workshop on Field Robotics	First Prize	Jun 2026
The Arthur and Louise May Memorial Fund Scholarship	First Prize	Jun 2025
Unmanned Aerial Vehicle Intelligent Perception Technology Competition	Second Prize	Nov 2022
ICRA RoboMaster AI Challenge	Champion	May 2022
RoboMaster University Sim2Real Challenge	Champion	May 2022
ICRA RoboMaster AI Challenge	Champion	Jun 2021
ICRA RoboMaster AI Challenge (China Division)	Second Prize	Mar 2021
RoboMaster University Championship	Second Prize	Aug 2020
Guangdong Provincial Electronic Design Contest ("5G+AI" Theme)	Third Prize	Aug 2020
National Undergraduate Electronic Design Contest (Guangdong Provincial Division)	Second Prize	Aug 2019
Freshman Annual Project Program in Harbin Institute of Technology, Shenzhen	First Prize	Jun 2019
2nd Youth Maker Marathon in Harbin Institute of Technology, Shenzhen	Second Prize	Mar 2019

Research

Ph.D. Candidate, the University of Hong Kong, Supervised by Prof. Fu Zhang Mar 2023 – Present

- **Integrated Planning and Control for Quadrotor Navigation:** In this work, we propose an integrated planning and control framework called IPC, consisting of a lightweight planning frontend and a linear MPC backend. The quadrotor is modeled as a high-order linear system, with obstacle avoidance enforced via linear hard constraints in MPC. The controller runs at 100 Hz with a computation time of 2–3.5 ms, enabling fast response to dynamic obstacles and improved disturbance rejection. The approach outperforms state-of-the-art methods in simulations with suddenly appearing obstacles and is validated on a fully autonomous LiDAR-based quadrotor in real-world dynamic environments. [Video]
- **Slope Inspection under Dense Vegetation Using LiDAR-based Quadrotors:** In this work, we develop a LiDAR-based quadrotor with a comprehensive software system. Our quadrotor features assisted obstacle avoidance, enabling it to autonomously avoid intricate obstacles while executing pilot commands. Field experiments conducted in collaboration with the Hong Kong Civil Engineering and Development Department demonstrate our quadrotor's ability to avoid small obstacles and maneuver in dense vegetation, validating its practical potential for slope inspection. [Video]
- **LiDAR-based Quadrotor Inspection System for Various Tasks in Cluttered Environments:** In this work, we built a complete LiDAR-based autonomous UAV inspection system. There are two phases in our system, human-in-the-loop inspection and autonomous inspection. In the first phase, even untrained pilots benefiting from autonomous obstacle avoidance, can control the UAV to effectively scan 3D maps, specify inspection points, and schedule tasks in cluttered and challenging environments. The inspection points are then optimized using the Traveling Salesman Problem (TSP) to develop efficient task sequences. In the second phase, the quadrotor autonomously executes the planned tasks, ensuring safe, efficient, and autonomous inspection in the same scenes. Comprehensive field experiments validate the system across various environments, including slopes, landslides, agricultural fields, factories, and forests, demonstrating its versatility and effectiveness in real-world inspection scenarios. [Video]
- **Obstacle-Avoidance-Assured UAV Flight amidst Uncertain Disturbances:** In this work, we propose a disturbance-aware integrated planning and control framework (DIPC) that jointly estimates multi-source disturbances using a Kalman filter and embeds them into a linear integrated planning and control model predictive control (MPC) problem, enabling safe trajectory generation and strong disturbance rejection

control at 100 Hz. Crucially, DIPC exhibits two adaptive capabilities to ensure obstacle avoidance under uncertain disturbances: (i) obstacle-aware disturbance rejection control, which adapts the disturbance rejection behaviors to obstacle avoidance, and (ii) disturbance-aware planning, which adapts planned paths to disturbance uncertainty. Wind tunnel experiments verify DIPC's capabilities, maintaining hovering errors below 10 cm in Beaufort scale 10 winds (25 m/s) and avoiding dynamic obstacles under 25 m/s steady winds, 20 m/s peak wind shear and 20 m/s peak sinusoidal winds. Real-world applications demonstrate DIPC's applications in close-proximity bridge inspection, near-ground terrain-following flight, high-speed flight, and varying-payload flight. These results represent a milestone in achieving obstacle-avoidance-assured UAV flight amidst uncertain disturbances. [Video]

- **Catapult Quadrotor Swarm System for Rapid Deployment:** In this work, our focus is on developing an inventive catapult quadrotor swarm system tailored to expedite deployment in emergency rescue scenarios. This system integrates foldable quadrotors, catapults, and a landing platform, all of which can be affixed to a mobile platform. As the mobile platform is moving, the catapult is capable of propelling the foldable quadrotors to the specified altitude. Subsequently, upon attaining flight, the foldable quadrotors will extend their arms, execute their designated flight tasks, and ultimately land on the landing platform situated on the mobile platform. This pioneering design is poised to significantly diminish the deployment duration of quadrotor swarms, thereby amplifying responsiveness and operational efficiency in emergency contexts.

Research Intern, The Chinese University of Hong Kong Research Institute (Futian) Jan 2021 – Jun 2021

- **Intelligent medicine box:** Developed the embedded system for an intelligent medicine box and implemented basic motion control of the mechanical components.

Project Experience

Critical-HIT Robotics Team, Harbin Institute of Technology, Shenzhen (HITSZ) Sep 2020 – Jun 2022

- **Robotics Team Captain:** Led the team's daily operations and project management, and contributed to hardware circuit design and embedded system development. Developed core functional modules including localization, navigation, autonomous aiming, and decision-making, for ground robotic platforms. [Video 2021] [Video 2022]
- **Autonomous UAV Advisor:** Supervised undergraduate teams in autonomous UAV racing competitions, with a focus on perception, state estimation, trajectory planning, and control for high-speed autonomous traversal through gates while avoiding static and dynamic obstacles. [Video]
- **Sentry Robot Advisor:** Provided technical guidance on full-stack system development, including mechanical design, control and planning algorithms, embedded hardware and software. Led the implementation of swerve-drive-based omnidirectional mobility, and the design of autonomous target search and tracking with automatic aiming in a 3D environment. [Video]

Industry Intern, SZ DJI Technology Co.,Ltd. Aug 2021 – Feb 2022

- **UAV Systems Engineer:** Conducted pre-research on quadrotor mobile landing technology, enabling autonomous landing on a moving ground vehicle at speeds up to 30 km/h.

National Undergraduate Electronic Design Contest, HITSZ Jul 2019 – Aug 2019

- **Team Captain (Problem H – Simulated Curved Electromagnetic Launcher):** Led team organization and task allocation, and was responsible for mechanical structure design and embedded system development.

Additional Experience And Skills

Teaching Assistant: MECH3433 Robotics, drones and autonomous ground vehicles (2025), MECH3418 Dynamics and control (2023-2024)

Interests: Table Tennis, Football, Robot Technology, Electronic Technology

Awards: 2014 Meizhou 7th City Sports Games: 4th Place (Men's Group A, Table Tennis – Team Event), Harbin Institute of Technology (Shenzhen) University-Level Sports Games: 3rd Place in Men's Team Events (Table Tennis and Football).